VIRENDRA DILIP MAHAJAN

AI Engineer | LLM Systems, RAG & Agentic AI

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PROFILE

Final-year B.Tech Computer Science student at MIT ADT University, Pune (2023–2027), specialising in LLM-based systems. Led a two-person team that built ExamOne, an AI-graded proctored examination platform now adopted by the university for roughly 7,000 students. Built the Planetary Compute Network, a distributed compute platform for ML workloads. Winner of the IEEE national-level AI Grand Challenge 2026. Three internships across full stack, data science and defence-tech research.

TECHNICAL SKILLS

AI Engineering: LLM fine-tuning (Qwen-14B), RAG pipelines, agentic AI, LangChain, LangGraph, prompt and system-prompt design, vector search (pgvector), embeddings, model evaluation

Languages: Python, JavaScript, SQL, SystemVerilog; working knowledge of Java and C++

Backend & Infrastructure: FastAPI, Node.js, PostgreSQL, REST APIs, Docker, sandboxed execution (Judge0), Oracle Cloud, distributed job orchestration, Linux

Frontend: React.js, Vite, HTML5, CSS3, Firebase

PROJECTS

ExamOne — AI-Graded Proctored Examination Platform | Team Lead, 2 engineers | React + Vite, Python, PostgreSQL + pgvector

- Delivered a complete examination platform handling MCQ, descriptive and coding papers in 16 days, and secured MIT ADT University as first client — going live for approximately 7,000 students from September 2026.
- Fine-tuned Qwen-14B on university syllabus material to grade descriptive answers, scoring each response against a reference answer in the question bank while reasoning independently rather than string-matching.
- Designed a Claude-based review layer that audits grader output and rewrites the model's system prompt, forming a self-correcting loop; grades were validated against faculty marking before rollout.
- Integrated an isolated Judge0 sandbox so untrusted student code executes safely and is scored automatically against test cases.
- Built anti-malpractice controls — enforced full-screen lockdown, disabled exit paths, keystroke logging and tab-switch detection — avoiding invasive camera surveillance.

Planetary Compute Network (PCN) — Distributed Compute Platform | planetarycomputenetwork.online

- Built a platform that pools ordinary connected computers into a single compute fabric for machine learning and deep learning workloads.
- Benchmarked close to 7x speedup over a single machine across 12 parallel workers; the system scales to 30 nodes.
- Designed a hybrid coordination model where nodes communicate peer-to-peer or via a central orchestrator that assigns work, tracks node health and aggregates results.
- Added virtual machine hosting on the pooled network, backed by Oracle Cloud infrastructure.

PCN-Semiconductors — Automated RISC-V Chip Design & Simulation | SystemVerilog

- Built a drag-and-drop design environment that auto-generates SystemVerilog from placed components, removing manual RTL authoring.
- Wrote a custom simulator testing generated chips under real-world conditions including thermal behaviour and current draw.
- Targets RISC-V designs for drones and electric vehicles; prototype complete, deployment pending.

Fasal Astra — Computer Vision Weed Detection | IEEE AI Grand Challenge 2026 — winning project

- Led development of a vision model identifying weeds in crop imagery to enable targeted spraying over blanket application; placed first against 40+ teams nationally.

Also on GitHub: Valhalla Suite (microservices platform — Docker, Node.js, PostgreSQL, RabbitMQ), AUTO AUTO (civic-tech PWA, deployed), Healthstride Manager, TrendTracker+.

EXPERIENCE

Technology Intern (Winter) — *Bharat Space Education Research Centre — Def-Space Tech* | Dec 2025 – Jan 2026

- Researched applications of emerging AI methods to defence and space-domain problems within a research team.

Data Science Intern (Virtual) — *SmartED Innovations* | Oct 2025 – Nov 2025

- Built Python data pipelines with Pandas and NumPy — cleaning, transforming and analysing raw datasets through to reported findings.

Full Stack Developer Intern — *Alpha Innovation Pvt Ltd* | Jun 2025 – Aug 2025

- Shipped frontend and backend features into live application modules using React.js, Node.js and JavaScript within an agile team.
- Debugged production issues and worked through standard code review, testing and version-control practices.

ACHIEVEMENTS

- Winner — IEEE National-Level AI Grand Challenge 2026, IEEE Pune & Maharashtra Section (AI for Agriculture). Led team to first place against 40+ teams; ₹1,00,000 prize.
- Sole student team to have software adopted institutionally by MIT ADT University.
- Completed three internships across three domains before final year.

EDUCATION

B.Tech, Computer Science and Engineering — MIT Art, Design and Technology University, Pune | 2023 – 2027
CGPA: 8.09 / 10 | Latest semester SGPA: 9.00
Relevant coursework: Deep Learning, Machine Learning, Artificial Intelligence, Data Structures & Algorithms, Container Based Application Development, Azure and GCP Essential Services.
Class 12 — 74.50% (2023) | Class 10 — 96.40% (2021)

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 — AI Foundations Associate • Oracle Cloud Infrastructure 2025 — Foundations Associate • Oracle Data Platform 2025 — Foundations Associate • AWS APAC Solutions Architecture Job Simulation (Forage)
Languages: English (professional), Hindi (professional), Marathi (native)